

Test of Hypothesis

Hypothesis is some sort of assumption on parameter. Whether the hypothesis is valid or not, true or not, justified or not, we will test it, this is called test of hypothesis. A vital role of Statistics is in verifying statements, claims and in general testing hypothesis. Based on a random sample, we can use Statistics to verify whether

- the average connection speed is 54 mbps, as claimed by the internet service provider,
- the average number of concurrent users increased by 2000 this year,
- the proportion of defective products is at most 3%, as promised by the manufacture,
- the number of errors in software is independent of the manager's experience, etc.

Suppose an internet server claims that computer users in a country spent on average 15 hours per week on browsing. To determine whether this was overstated, a competitor conducted a survey of 250 users. If we are interested in arriving at decision on whether the claim of the internet server is justified, the problem falls in the area of hypothesis testing.

A statistical hypothesis is a statement or assumption about parameter. A hypothesis is denoted by H .

Null hypothesis: The word null means the possible rejection of the hypothesis. The null hypothesis, denoted by H_0 , is the hypothesis that we want to reject. That is, a null hypothesis is a statistical hypothesis that is stated for possible rejection under the assumption that it is true.

Alternative hypothesis: The hypothesis contrary to the null hypothesis under the test is called alternative hypothesis. An alternative hypothesis is denoted by H_1 . Let $x_1, x_2, \dots, x_n$ be a random sample from $f(x, \theta)$. We want to test:

$$H_0: \theta = \theta_0$$

$$\text{Vs } H_1: \theta \neq \theta_0$$

where θ_0 is the specified value or null value.

Sample space and Critical/ Rejection region: Suppose $x_1, x_2, \dots, x_n$ be the sample observations. All the values of sample observations will be the aggregate of a sample and they constitute a space, called sample space which is denoted by S .

If, we divide the sample space in two mutually exclusive regions, say w and $\bar{w}$ [$S - w$], the null hypothesis is rejected if the observed sample points fall in w and it is accepted if the sample point fall in $\bar{w}$. Hence, w is the critical or rejection region and $\bar{w}$ is the acceptance region.

One tailed and two tailed test: The test for which the critical region is located at both ends of the sampling distribution is called a two tailed test. The following hypothesis,

$H_0: \mu = \mu_0$ Vs $H_1: \mu \neq \mu_0$ is an example of two tailed test.

The test for which the critical region is located at one end (either left or right) of sampling distribution is called a one tailed test. The following hypothesis,

$H_0: \mu = \mu_0$ Vs $H_1: \mu > \mu_0$ is an example of one tailed test (right tail).

$H_0: \mu = \mu_0$ Vs $H_1: \mu < \mu_0$ is an example of one tailed test (left tail).

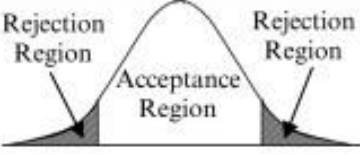
One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$	$H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$	$H_0 : \mu = \mu_0$ $H_1 : \mu > \mu_0$
		

Figure: Acceptance and rejection/critical region for one tailed test and two tailed test.

Examples:

1. To verify that the average connection speed is 54 Mbps, we test the hypothesis $H_0 : \mu = 54$ against the two-sided alternative $H_1 : \mu \neq 54$, where μ is the average speed of all connections. However, if we worry about a low connection speed only, we can conduct a one-sided test of

$H_0 : \mu = 54$ vs $H_1 : \mu < 54$. In this case, we only measure the amount of evidence supporting the one-sided alternative $H_1 : \mu < 54$. In the absence of such evidence, we gladly accept the null hypothesis.

2. To verify if the proportion of defective products is at most 3%, we test $H_0 : p = 0.03$ vs $H_1 : p < 0.03$, where p is the proportion of defects in the whole shipment. We choose the left-tailed alternative $H_1 : p < 0.03$ because we will not reject the shipment only if significant evidence supporting this alternative is collected.

3. To verify whether the average number of concurrent users is increased, we test $H_0 : \mu_2 - \mu_1 = 0$ vs $H_1 : \mu_2 - \mu_1 > 0$.

Type I and Type II error: In hypothesis testing, one of the following four situations will occur:

	H_0	True	False
Decision	Accept	No error	Type II error (β)
	Reject	Type I error (α)	No error

A type I error occurs when we reject the true null hypothesis. A type II error occurs when we accept the false null hypothesis. Each error occurs with a certain probability that we hope to keep small. A type I error is often considered more dangerous and undesired than a type II error. Making a type I error can be compared with convicting an innocent defendant or sending a patient to a surgery when (s)he does not need one. For this reason, we shall design tests that bound the probability of type I error by a pre assigned small number α . Under this condition, we may want to minimize the probability of type II error.

Level of significance:

Probability of a type I error is the significance level of a test which is denoted by α . That is,

$$\alpha = P \{ \text{reject } H_0 \mid H_0 \text{ is true} \}.$$

Typically, hypotheses are tested at significance levels as small as 0.01, 0.05, or 0.10, although there are exceptions. Testing at a low level of significance means that only a large amount of evidence can force rejection of H_0 . Rejecting a hypothesis at a very low level of significance is done with a lot of confidence that this decision is right.

Test statistic:

A test statistic is a numerical value calculated from sample data that is used to decide whether to reject the null hypothesis H_0 . It measures how far the sample data deviate from what is expected under H_0 . Mathematically,

$$\text{Test statistic} = \frac{\text{Observed value} - \text{Expected value Under } H_0}{\text{Standard Error}}$$

Important Tests

The important test of significance in statistics can be classified as follows:

1. Normal test
2. t test
3. χ^2 test
4. F test

Normal test:

In normal tests we find a statistic u whose expected value $E(u)$ is specified by the null hypothesis. The standard error $\sigma(u)$ of u is either known or estimated from large sample ($n > 30$). Then,

$$z = \frac{u - E(u)}{\sigma(u)} \sim N(0, 1)$$

If the distribution of u is normal and $\sigma(u)$ is known, the statistic z is exactly normally distributed and z is approximately normal if either u is normal or $\sigma(u)$ is estimated from a large sample or both. This test is called large sample test.

t test:

In normal test it is assumed that $\sigma(u)$ is known or estimated from large sample ($n > 30$). But when the sample size is small ($n \leq 30$) and $\sigma(u)$ is usually unknown, W.S. Gosset derived a test statistic t and obtained its sampling distribution to meet this situation. This statistic is called student

t statistic and its sampling distribution is called student t distribution. Thus

$$t = \frac{u - E(u)}{\sigma(u)} \sim t_v \text{ with } v \text{ degrees of freedom(d.f.)}$$

where v is called the degrees of freedom(d.f.) of the distribution. This test is called small sample test.

 χ^2 test:

χ^2 test are used mainly for testing the hypothesis that specify the nature of one or more distributions as a whole. It may also specify that certain attributes are independent. Let $O_i (i = 1, 2, 3, \dots, k)$ be the observed frequency of the i th class and E_i be the expected frequency of the same class. Then, the test statistic χ^2 is defined as

$$\frac{\sum_{i=1}^k (O_i - E_i)^2}{E_i} \sim \chi_v^2 \text{ with } v \text{ degrees of freedom(d.f.)}$$

F test:

The test was originally derived by R.A. Fisher and Snedecor called it F test in honor of Fisher. Let S_1^2 and S_2^2 are two independent estimates of variance of normal population. Then statistic F is defined as

$$F = \frac{S_1^2}{S_2^2} \sim F_{v_1, v_2} \text{ with } v_1 \text{ and } v_2 \text{ degrees of freedom(d.f.)}$$

The test statistic F is so defined that the estimate S_1^2 in the numerator of F is expected to be larger than S_2^2 when the null hypothesis is not true. Hence only upper tail of F distribution is used as critical region.

Test of Significance about Mean

Here, we consider the following:

1. Comparison of a sample mean with an assigned population mean
2. Comparison of two **independent** sample means
3. Comparison of two **correlated** sample means

1. Comparison of a sample mean with an assigned population mean

Suppose that, we have a random sample of size n from a normal population having mean μ and variance σ^2 . We want to test the hypothesis that the population mean μ is equal to some assigned value, say μ_0 .

Depending on whether the population standard deviation (σ) is known or unknown; if unknown, whether it is estimated from large or small sample, the following cases arise-

- a. σ is assumed to be known or estimated from a large sample
- b. σ is unknown and estimated from a small sample

a. σ is assumed to be known or estimated from a large sample:

Sometimes the population variance σ^2 is known; more often we can derive a good estimate s^2 of σ^2 from an observed sample that is assumed to be large. In both situations we use the standard normal test or z -test. That is,

We are to test-

$H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$

Assumptions:

- i) The sample is taken randomly from a normal population/distribution having mean μ and variance σ^2 .
- ii) The standard deviation σ is known or estimated from large sample ($n \geq 30$, say).

The test statistic is-

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \text{ [for known } \sigma]$$

$$\text{or, } z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim N(0,1) \text{ [for unknown } \sigma \text{ but estimated from large sample } (n \geq 30)]$$

$$\text{Where, } s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{n-1} [\sum x_i^2 - n\bar{x}^2]$$

Let, the level of significance = α

Since, the test is a two tailed test, the critical values of the test statistic are $z_{\alpha/2}$ and $z_{1-\alpha/2}$.

Comment: If $z < z_{\alpha/2}$ or $z > z_{1-\alpha/2}$ (i.e., z falls in the rejection area), we may reject the null hypothesis. Finally, we conclude that, the population mean is not equal to μ_0 .

95% Confidence Interval (CI) for μ is

$$-1.96 \leq \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq 1.96$$

$$\Rightarrow \bar{x} - 1.96 \sigma / \sqrt{n} \leq \mu \leq \bar{x} + 1.96 \sigma / \sqrt{n}$$

Decision rule for one-tailed test:

For right tailed test: $H_0: \mu = \mu_0$ against $H_1: \mu > \mu_0$

Reject H_0 if $z > z_{1-\alpha}$

For left tailed test: $H_0: \mu = \mu_0$ against $H_1: \mu < \mu_0$

Reject H_0 if $z < z_\alpha$

A table to decision rule for z-test

	Decision rule	
H_1	$\alpha = .05$	$\alpha = .01$
	Reject H_0 if and only if	Reject H_0 if and only if
$\mu > \mu_0$	$z > z_{1-\alpha}$ where $z_{1-\alpha} = 1.64$	$z > z_{1-\alpha}$ where $z_{1-\alpha} = 2.33$
$\mu < \mu_0$	$z < z_\alpha$ where $z_\alpha = -1.64$	$z < z_\alpha$ where $z_\alpha = -2.33$
$\mu \neq \mu_0$	$z > z_{1-\alpha/2} \& z < z_{\alpha/2}$ where $z_{1-\alpha/2} = 1.96$ and $z_{\alpha/2} = -1.96$	$z > z_{1-\alpha/2} \& z < z_{\alpha/2}$ where $z_{1-\alpha/2} = 2.58$ and $z_{\alpha/2} = -2.58$

Example 1: A bulb manufacturing company claims that the average longevity of their bulb is 3 years. It is given that the variance of longevity of manufactured bulb

is 0.5 years. A random sample of 35 bulbs gave a mean longevity of 2.6 years. Does the sample mean justify the claim of the manufacturer? Test this at 5% level of significance. Also find 95% confidence interval for population mean.

Solution:

Hypothesis: $H_0: \mu = 3$ against $H_1: \mu \neq 3$

Level of significance: $\alpha = .05$

This is a two tailed test. The critical values for this test are, $z_{\frac{\alpha}{2}} = z_{.025} = -1.96$ and $z_{1-\frac{\alpha}{2}} = z_{.975} = 1.96$.

Critical/Rejection region: $z < -1.96$ or $z > 1.96$

Test statistic: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$; where, $\bar{x} = 2.6$, $\sigma = \sqrt{.5} = 0.707$, $\mu_0 = 3$, $n = 35$.

Calculated value of test statistic: $z = \frac{2.6-3}{.707/\sqrt{35}} = -3.34664$.

Comparison and decision: Since, calculated value of z falls in the critical/ rejection region, we may reject null hypothesis at 5% level of significance.

Comment: The result implies that the claim of the bulb manufacturing company is false. That is, the mean longevity of their bulb is not 3 years.

95% Confidence Interval (CI) for μ is

$$\bar{x} - 1.96 \sigma / \sqrt{n} \leq \mu \leq \bar{x} + 1.96 \sigma / \sqrt{n}$$

$$\Rightarrow 2.6 - 1.96 \cdot 0.707 / \sqrt{35} \leq \mu \leq 2.6 + 1.96 \cdot 0.707 / \sqrt{35}$$

$$\Rightarrow 2.3658 \leq \mu \leq 2.8342$$

Another way to take decision: Use of p-value

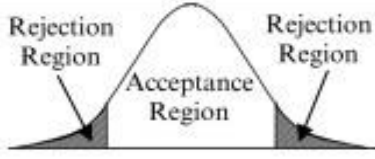
p-value:

The p-value or the probability value is the the probability of obtaining a test statistic as extreme as (or more extreme than) the observed value, assuming the null hypothesis is true.

Decision rule:

- If $p \leq \alpha \rightarrow$ Reject H_0
- If $p > \alpha \rightarrow$ Fail to reject H_0 .

Where α is the significance level (0.10, 0.05, 0.01)

One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$	$H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$	$H_0 : \mu = \mu_0$ $H_1 : \mu > \mu_0$
		
$p - value$ $= P[Z \leq Z_{cal}]$	$p - value$ $= P[Z \leq - Z_{cal} \text{ or } Z \geq Z_{cal}]$ $= P[Z \leq - Z_{cal}] + P[Z \geq Z_{cal}]$ $= 2 \times P[Z \geq Z_{cal}]$ $= 2 \times (1 - P[Z \leq Z_{cal}])$	$p - value = P[Z \geq Z_{cal}]$ $= 1 - P[Z \leq Z_{cal}]$

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For the above problem (Ex-2), test statistic: $z_{cal} = \frac{2.6-3}{.707/\sqrt{35}} = -3.34664$, and it is

two-tailed test. So, $p - value = 2 \times (1 - P[Z \leq |Z_{cal}|]) = 2 \times (1 - P[Z \leq |-3.35|])$

$= 2 \times (1 - P[Z \leq 3.35]) = 2 \times (1 - 0.9996) = 0.0008$. [Using Z table]

Therefore, $p - value = 0.0008$.

Decision: We consider $\alpha = 0.05$. As $p\text{-value} < \alpha$, we may reject H_0 .

Example 2: The average IQ of university women in Bangladesh is suspected to be 110. A random sample of 64 women yielded an average IQ of 115.5 and a standard deviation of 20. Can you conclude the average score of the women in the population is really more than 110? Test this at 1% level of significance.

Solution:

Hypothesis: $H_0: \mu = 110$ against $H_1: \mu > 110$

Level of significance: $\alpha = .01$

This is a one tailed test. The critical value for this test is, $z_{1-\alpha} = z_{.99} = 2.33$.

Critical/Rejection region: $z > 2.33$

Test statistic: $z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$; where, $\bar{x} = 115.5$, $s = 20$, $\mu_0 = 110$, $n = 64$.

Calculated value of test statistic: $z = \frac{115.5-110}{20/\sqrt{64}} = 2.2$.

Comparison and decision: Since, calculated value of z falls in the acceptance region, we may not reject null hypothesis at 1% level of significance.

Comment: The result implies the average IQ score of women is not greater than 110.

Exercise 1: The strength of steel wire made by an existing process is normally distributed with a mean of 1250 and a standard deviation of 150. A batch of wire is made by a new process, and a random sample consisting of 25 measurements gives an average strength of 1312. Assume that the standard deviation does not change. Is there evidence at the 1% level of significance that the new process gives a larger mean strength than the old? [Hint: $H_0: \mu = 1250$ against $H_1: \mu > 1250$]

Exercise 2: The average daily amount of scrap from a particular manufacturing process is 25.5 kg with a standard deviation of 1.6 kg. A modification of the process is tried an attempt to reduce this amount. During a 10-day trial period, the kilograms of scrap produced each day were: 25.0, 21.9, 23.5, 25.2, 22.0, 23.0, 24.5, 25.0, 26.1, 22.8. From the nature of the modification, no change in day-to-day variability of the amount of scrap will result. The normal distribution will apply. A first glance at the figures suggests that the modification is effective in reducing the scrap level. Does a significance test confirm this at the 1% level? [Hint: $H_0: \mu = 25.5$ against $H_1: \mu < 25.5$]

Exercise 3: The number of concurrent users for some internet service provider has always averaged 5000 with a standard deviation of 800. After an equipment upgrade, the average number of users at 100 randomly selected moments of time is 5200. Does it indicate, at a 5% level of significance, that the mean number of concurrent users has increased? Assume that the standard deviation of the number of concurrent users has not changed. [Hint: $H_0: \mu = 5000$ against $H_1: \mu > 5000$]

Exercise 4: Suppose that the steel manufacturing company wishes to know whether the tensile strength of the steel wire has an overall average of 120 pounds. A sample of 25 units of steel wire produced by the company yields a mean strength of 110 pounds and a variance of 144 pounds². Should the company conclude that the average strength is not 120 pounds with $\alpha=.05$?

b. σ is unknown and estimated from a small sample:

The population variance is usually unknown and the sample size is often small ($n < 30$). In these situations, we use t test. That is,

We are to test-

$$H_0: \mu = \mu_0 \text{ against } H_1: \mu \neq \mu_0$$

Assumptions:

- i) The sample is taken randomly from a normal population/distribution having mean μ and variance σ^2 .
- ii) The standard deviation σ is unknown and estimated from small sample ($n < 30$).

The test statistic is-

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ with } (n-1) \text{ degrees of freedom.}$$

$$\text{Where, } s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{n-1} [\sum x_i^2 - n\bar{x}^2]$$

Let, the level of significance $= \alpha$

Since, the test is a two tailed test, the critical values of the test statistic are $\pm t_{\frac{\alpha}{2}, n-1}$

Comment: If $t > t_{\frac{\alpha}{2}, n-1}$ or $t < -t_{\frac{\alpha}{2}, n-1}$, we may reject the null hypothesis. Finally, we conclude that, the population mean is not equal to μ_0 .

100(1 - α)% Confidence Interval (CI) for μ is

$$-t_{\frac{\alpha}{2}, n-1} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}} \leq t_{\frac{\alpha}{2}, n-1}$$

$$\Rightarrow \bar{x} - t_{\frac{\alpha}{2}, n-1} S/\sqrt{n} \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}, n-1} S/\sqrt{n}$$

Decision rule for one-tailed test:

For right tailed test: $H_0: \mu = \mu_0$ against $H_1: \mu > \mu_0$

Reject H_0 if $t > t_{\alpha, n-1}$

For left tailed test: $H_0: \mu = \mu_0$ against $H_1: \mu < \mu_0$

Reject H_0 if $t < -t_{\alpha, n-1}$

Example 1: It is stated by a company that the daily average production (in terms of number) of the workers is 20. A sample of 10 workers is taken from that company and their productions are as: 40, 35, 48, 43, 35, 39, 44, 37, 42, 38. At 5% level of significance test the statement of the company assuming the production follows normal distribution.

Solution: Hypothesis: $H_0: \mu = 20$ against $H_1: \mu \neq 20$

Level of significance: $\alpha = .05$

This is a two tailed test. The critical values for this test are, $t_{\frac{\alpha}{2}, 9} = t_{0.025, 9} = 2.262$

Critical/Rejection region: $t < -2.262$ or $t > 2.262$

Test statistic: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$; where, $\bar{x} = 40.1$, $s = \sqrt{\frac{1}{n-1} [\sum x_i^2 - n\bar{x}^2]} = 4.175$, $\mu_0 = 20$, $n = 10$.

Calculated value of test statistic: $t = \frac{40.1 - 20}{4.175/\sqrt{10}} = 15.224$.

Comparison and decision: Since, calculated value of t falls in the critical/ rejection region, we may reject null hypothesis at 5% level of significance.

Comment: The result implies that the claim of the company is false. That is, daily average production (in terms of number) of the workers is not 20.

Comparison of Two Independent Sample Means

Suppose that, we have two independent random samples of sizes n_1 and n_2 respectively from two populations. These populations are assumed to be normal with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively.

Let, $\bar{x}_1$ and $\bar{x}_2$ are the sample means and s_1^2 and s_2^2 are the sample variances.

We want to test whether the population means are equal or not. That is, $H_0: \mu_1 = \mu_2$.

Since, the test statistic depends on some assumptions like whether the variances σ_1^2 and σ_2^2 are known or not, whether the variances σ_1^2 and σ_2^2 are equal or not and whether the samples are large or small, here we will consider the following three cases:

- a. σ_1^2 and σ_2^2 are known
- b. σ_1^2 and σ_2^2 are unknown but equal
 - i) Samples are large
 - ii) Samples are small
- c. σ_1^2 and σ_2^2 are unknown and unequal

- i) Samples are large
- ii) Samples are small

a. σ_1^2 and σ_2^2 are known:

We are to test-

$$H_0: \mu_1 = \mu_2 \text{ or } H_0: \mu_1 - \mu_2 = 0$$

against $H_1: \mu_1 \neq \mu_2$

Assumptions:

- i) The samples are taken from two independent normal populations.
- ii) σ_1^2 and σ_2^2 are known.

Here, $(\bar{x}_1 - \bar{x}_2)$ is our relevant statistic.

$$V(\bar{x}_1 - \bar{x}_2) = V(\bar{x}_1) + V(\bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Therefore, the test statistic,

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - E(\bar{x}_1 - \bar{x}_2)}{\sqrt{V(\bar{x}_1 - \bar{x}_2)}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1) \text{ [for known } \sigma_1^2 \text{ and } \sigma_2^2]$$

Let, the level of significance = α

Since, the test is a two tailed test, the critical values of the test statistic are $z_{\alpha/2}$ and $z_{1-\alpha/2}$.

Comment: If $z > z_{1-\alpha/2}$ or $z < z_{\alpha/2}$, we may reject the null hypothesis. Finally, we conclude that, the population means are not equal (i.e. $\mu_1 \neq \mu_2$).

Decision rule for one-tailed test:

For right tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$

Reject H_0 if $z > z_{1-\alpha}$

For left tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 < \mu_2$

Reject H_0 if $z < z_\alpha$

Problem 1: Two-Tailed Test

Two independent samples are taken from two normal populations.

$$n_1 = 50, \bar{X}_1 = 105, \sigma_1 = 10$$

$$n_2 = 60, \bar{X}_2 = 100, \sigma_2 = 12$$

Test at 5% level whether the two population means are equal.

Solution

Step 1: Hypotheses

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step 2: Test statistic

$$\begin{aligned} Z &= \frac{105 - 100}{\sqrt{\frac{10^2}{50} + \frac{12^2}{60}}} \\ &= \frac{5}{\sqrt{\frac{100}{50} + \frac{144}{60}}} &= \frac{5}{2.098} \\ & & Z = 2.38 \end{aligned}$$

Step 3: Critical value

At 5% level (two-tailed):

$$Z_{0.025} = 1.96$$

Step 4: Decision

Since $2.38 > 1.96$, reject H_0 .

Conclusion: The two population means are significantly different.

Problem: Suppose we want to see if there is any difference between the average number of children born to the tribal women and the non-tribal women of a certain administrative division in Bangladesh. A random sample of 12 women from tribal population and 15 women from non-tribal population were drawn and their number of children recorded. The average number of children born to these two groups of women were 4.5 and 3.4 respectively. The population variances were 1 and 1.5 respectively. Use 5 percent level of significance to see if the sample data reflect any difference in the mean number of children born in the population.

Solution:

Hypothesis: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$

Level of significance, $\alpha = 0.05$

The test statistic,

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

Where, $\bar{x}_1 = 4.5$, $\bar{x}_2 = 3.4$, $\sigma_1^2 = 1$, $\sigma_2^2 = 1.5$, $n_1 = 12$, $n_2 = 15$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{4.5 - 3.4}{\sqrt{\frac{1}{12} + \frac{1.5}{15}}} = 2.57$$

Critical region: $z < -1.96$ or $z > 1.96$

Conclusion: Since the calculated value of z falls in the critical region, we may reject null hypothesis at 5% level of significance. So we can conclude that there is difference between the true average number of children born to tribal women and non-tribal women.

Exercise. A company claims that Machine A produces items with higher average weight than Machine B.

$$n_1 = 40, \bar{X}_1 = 52, \sigma_1 = 8$$

$$n_2 = 45, \bar{X}_2 = 49, \sigma_2 = 6$$

Test at 1% level whether Machine A has higher mean weight.

b. when σ_1^2 and σ_2^2 are unknown but equal:

We are to test-

$$H_0: \mu_1 = \mu_2 \text{ or, } H_0: \mu_1 - \mu_2 = 0$$

against $H_1: \mu_1 \neq \mu_2$

Assumptions:

- i) The samples are taken from two independent normal populations.
- ii) σ_1^2 and σ_2^2 are unknown but equal.

Here, $(\bar{x}_1 - \bar{x}_2)$ is our relevant statistic.

Let $\sigma_1^2 = \sigma_2^2 = \sigma^2$. Now

$$V(\bar{x}_1 - \bar{x}_2) = V(\bar{x}_1) - V(\bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2} = \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$$

The pooled estimate of σ^2 is-

$$S_{pooled}^2 = \frac{\sum(x_{1i} - \bar{x}_1)^2 + \sum(x_{2i} - \bar{x}_2)^2}{(n_1 - 1) + (n_2 - 1)} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)} = s^2 \text{ (Say)}$$

$$\text{So, } V(\bar{x}_1 - \bar{x}_2) = s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)$$

i) When samples are large (that is, $n_1 \geq 30$ and $n_2 \geq 30$)

The test statistic,

$$z = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0, 1)$$

Let, the level of significance = α

Since, the test is a two tailed test, the critical values of the test statistic are $z_{\alpha/2}$ and $z_{1-\alpha/2}$.

Comment: If $z > z_{1-\alpha/2}$ or $z < z_{\alpha/2}$, we may reject the null hypothesis. Finally, we conclude that, the population means are not equal (i.e. $\mu_1 \neq \mu_2$).

Decision rule for one-tailed test:

For right tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$

Reject H_0 if $z > z_{1-\alpha}$

For left tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 < \mu_2$

Reject H_0 if $z < z_{\alpha}$

Problem 1: Two-Tailed Test

Two large samples are drawn:

$$n_1 = 40, \bar{X}_1 = 52, s_1^2 = 25$$

$$n_2 = 50, \bar{X}_2 = 48, s_2^2 = 36$$

Test at 5% level whether population means differ. Assume equality of variances in the population.

Solution

Step 1: Hypotheses

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

The test statistic,

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1), \text{ where}$$

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

$$s = \sqrt{\frac{(40 - 1)25 + (50 - 1)36}{40 + 50 - 2}}$$

$$s = 5.58$$

So,

$$z = \frac{52 - 48}{5.58\sqrt{\left(\frac{1}{40} + \frac{1}{50}\right)}}$$

$$z = 3.38$$

At 5% level, critical value = 1.96

Since $3.38 > 1.96$, reject H_0 .

Conclusion: The means are significantly different.

Exercise 1. A university wants to compare two teaching methods for a statistics course. Method A was used for 45 students. Their average exam score was 74 marks, and the sample variance was 81. Method B was used for 50 students. Their average exam score was 70 marks, and the sample variance was 100. Assuming the population variances are equal, test at the 5% level whether there is a significant difference between the mean scores of the two methods.

ii) When samples are small (that is, $n_1 < 30$ and/or $n_2 < 30$)

The test statistic,

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1+n_2-2} \text{ with } n_1+n_2-2 \text{ degrees of freedom.}$$

Let, the level of significance = α

Since, the test is a two tailed test, the critical values of the test statistic are $\pm t_{\alpha/2, n_1+n_2-2}$

Comment: If $t < -t_{\alpha/2, n_1+n_2-2}$ or $t > t_{\alpha/2, n_1+n_2-2}$, we may reject the null hypothesis. Finally, we conclude that, the population means are not equal (i.e. $\mu_1 \neq \mu_2$).

Decision rule for one-tailed test:

For right tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$

Reject H_0 if $t > t_{\alpha, n_1+n_2-2}$

For left tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 < \mu_2$

Reject H_0 if $t < -t_{\alpha, n_1+n_2-2}$

Problem: An employer in a garments factory argues that it is justified to pay male workers more than female workers on the ground that they are more productive. Can the employer's claim be attained at the 5% level if the weekly output of a sample of 15 workers produced a mean of 350 men's wear with a standard deviation of 18 and the weekly outputs of 10 female workers produced a mean of 345 such wears with a standard deviation of 21? Assume equality of variance in the population.

Solution:

Hypothesis: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$

Level of significance, $\alpha = 0.05$

The test statistic,

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1+n_2-2} \text{ with } n_1+n_2-2 \text{ degrees of freedom.}$$

Where, $\bar{X}_1 = 350, \bar{X}_2 = 345, S_1 = 18, S_2 = 21, n_1 = 15, n_2 = 10$

$$\text{Now, } t_{cal} = \frac{350-345}{19.23 * \sqrt{\left(\frac{1}{15} + \frac{1}{10}\right)}} = 0.63$$

$$\text{Critical value, } t_{\alpha, n_1+n_2-2} = t_{0.05,23} = 1.714$$

Critical region: $t > 1.714$

Conclusion: Since the calculated value of t falls in the acceptance region, we may not reject null hypothesis. That is the difference between sample means may not be statistically significant. Hence the claim of the employer is not justified.

Problem 1. Two different medicines are tested to reduce blood pressure. Medicine A was given to 12 patients. Mean reduction in blood pressure is 8 mmHg and sample variance is 9. Medicine B was given to 10 patients. Mean reduction is 5 mmHg and sample variance is 16. Assuming equal population variances and normality, test at the 5% level whether the two medicines differ in their mean effectiveness.

Solution:

Hints, use t-test. $t=2.01$,

At 5% level, two-tailed,

$$t_{0.025,20} = 2.086$$

As, $2.01 < 2.086$, we fail to reject H_0 . There is no significant difference at the 5% level between the two medicines in reducing blood pressure.

c. when σ_1^2 and σ_2^2 are unknown and unequal:

We are to test-

$H_0: \mu_1 = \mu_2$ or, $H_0: \mu_1 - \mu_2 = 0$ against $H_1: \mu_1 \neq \mu_2$

Assumptions:

- i) The samples are taken from two independent normal populations.
- ii) σ_1^2 and σ_2^2 are unknown and unequal.

In this case,

$$V(\bar{x}_1 - \bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}$$

Where, s_1^2 and s_2^2 are two estimates of population variances σ_1^2 and σ_2^2 .

- i) When samples are large ($n_1 \geq 30$ and $n_2 \geq 30$)

The test statistic,

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \sim N(0,1)$$

Let, the level of significance = α

Since, the test is a two tailed test, the critical values of the test statistic are $z_{\alpha/2}$ and $z_{1-\alpha/2}$.

Comment: If $z > z_{1-\alpha/2}$ or $z < z_{\alpha/2}$, we may reject the null hypothesis. Finally, we conclude that, the population means are not equal (i.e. $\mu_1 \neq \mu_2$).

Decision rule for one-tailed test:

For right tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$

Reject H_0 if $z > z_{1-\alpha}$

For left tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 < \mu_2$

Reject H_0 if $z < z_\alpha$

Problem 1. A chemical plant is testing two different heat exchanger designs. For Design A, 45 units were tested. The average thermal efficiency was 82 percent. The sample variance of efficiency was 36. For Design B, 50 units were tested. The average thermal efficiency was 79 percent. The sample variance was 49. Test at the 5 percent level whether there is a significant difference between the average efficiencies of the two designs.

Problem 2. Two production lines manufacture the same polymer material. From Production Line 1, 60 samples were tested. The average tensile strength was 520 MPa. The sample variance was 100. From Production Line 2, 55 samples were tested. The average tensile strength was 505 MPa. The sample variance was 144. At the 1 percent level of significance, test whether Production Line 1 produces polymer with higher average tensile strength.

ii) When samples are small ($n_1 < 30$ and/or $n_2 < 30$)

The test statistic,

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim t_v \text{ with } v \text{ degrees of freedom.}$$

$$\text{Where, } v = \frac{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})^2}{\frac{(\frac{s_1^2}{n_1})^2}{n_1 - 1} + \frac{(\frac{s_2^2}{n_2})^2}{n_2 - 1}}$$

Let, the level of significance = α

Since, the test is a two tailed test, the critical values of the test statistic are $\pm t_{\alpha/2, v}$

Comment: If $t < -t_{\alpha/2, v}$ or $> t_{\alpha/2, v}$, we may reject the null hypothesis. Finally, we conclude that, the population means are not equal (i.e. $\mu_1 \neq \mu_2$).

For right tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 > \mu_2$

Reject H_0 if $t > t_{\alpha, v}$

For left tailed test: $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 < \mu_2$

Reject H_0 if $t < -t_{\alpha, v}$

Problem 1. Two different packing materials are tested in a laboratory-scale distillation column. For Packing Material A, 12 experimental runs were conducted. The average separation efficiency was 78 percent. The sample variance was 16. For Packing Material B, 10 experimental runs were conducted. The average separation efficiency was 73 percent. The sample variance was 25. Assuming the efficiencies follow a normal distribution and the population variances are unequal, test at the 5 percent level whether the two packing materials differ in mean efficiency.

Problem 2. Two different catalysts are tested for a chemical reaction. Catalyst A was tested in 8 experimental runs. The average conversion rate was 85 percent. The sample variance was 49. Catalyst B was tested in 10 experimental runs. The average conversion rate was 78 percent. The sample variance was 100. Assuming the data are normally distributed and the population variances are not equal, test at the 5 percent level whether there is a significant difference in the mean conversion rates.

Problem 3. Two transformer core materials are being compared. Material A was tested on 10 transformers. The average temperature rise under full load was 65 degrees Celsius. The sample variance was 25. Material B was tested on 8 transformers. The average temperature rise was 72 degrees Celsius. The sample variance was 49. Assuming normal populations and unequal variances, test at the 5 percent level whether Material A results in lower mean temperature rise.

Comparison of Two Correlated Sample Means

In certain situations the two sample means under comparison are related to each other because the observations occur in pairs. Thus, instead of having two random samples, we have in effect one random sample of pairs. Each pair of observation is often obtained from the same individual.

For example, to test the effectiveness of a weight reducing diet, we may collect the weights of a set of patients before the experiment starts and after the experiment completed. In such situations, we use paired t test.

Suppose, we have n independent pairs of observations (x_i, y_i) , $i = 1, 2, \dots, n$.

We want to test the hypothesis-

$H_0: \mu_x = \mu_y$ or we can also form it as, $H_0: \mu_d = \mu_x - \mu_y = 0$

$H_1: \mu_d \neq 0$ where, $d_i = x_i - y_i$

Assumption:

$d_1, d_2, \dots, d_n$ constitute a random sample from a normal population $N(\mu_d, \sigma_d)$.

To test the hypothesis, the test statistic is-

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \sim t_{n-1} \text{ with } (n-1) \text{ degrees of freedom.}$$

Let, the level of significance, α

Since, the test is a two tailed test, the critical values of the test statistic are $\pm t_{\frac{\alpha}{2}, n-1}$.

Comment: If $t < -t_{\frac{\alpha}{2}, n-1}$ or $t > t_{\frac{\alpha}{2}, n-1}$ that is, we may reject the null hypothesis.

Finally, we conclude that, the population means are not equal (i.e. $\mu_x \neq \mu_y$).

For right tailed test: $H_0: \mu_d = 0$ against $H_1: \mu_d > 0$

Reject H_0 if $t > t_{\alpha, n-1}$

For left tailed test: $H_0: \mu_d = 0$ against $H_1: \mu_d < 0$

Reject H_0 if $t < t_{\alpha, n-1}$

Problem: The following data relate to the annual sales data in million taka for 2014 and 2015 from 10 randomly selected companies in Dhaka city. Test at 5% level that there is no difference between the two years sales record.

Company	A	B	C	D	E	F	G	H	I	J
Sales 2014	126	56	86	62	96	36	52	50	35	53
Sales 2015	123	49	79	59	92	35	48	48	32	50

Solution:

Hypothesis: $H_0: \mu_x - \mu_y = 0$ or, $H_0: \mu_d = 0$ against $H_1: \mu_d \neq 0$; where, μ_x and μ_y are average sales (in million taka) in Dhaka city for the year 2014 and 2015, respectively.

Level of significance, $\alpha = 0.05$

Test statistic: $t = \frac{\bar{d}}{s_d / \sqrt{n}} \sim t_{n-1}$

To carry out the test, let us compute the mean and standard deviation of the differences:

Company	A	B	C	D	E	F	G	H	I	J
Sales 2014 (x)	126	56	86	62	96	36	52	50	35	53
Sales 2015 (y)	123	49	79	59	92	35	48	48	32	50
d_i	3	7	7	3	4	1	4	2	3	3
d_i^2	9	49	49	9	16	1	16	4	9	9

$$\bar{d} = \frac{\sum d_i}{n} = \frac{37}{10} = 3.7, S_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}} = \sqrt{\frac{171 - 10*(3.7)^2}{9}} = 1.95$$

$$\text{Now, } t_{cal} = \frac{3.7}{1.95/\sqrt{10}} = 6.0$$

$$\text{Critical values, } \pm t_{\frac{\alpha}{2}, n-1} = \pm t_{0.025, 9} = \pm 2.262$$

Critical region: $t < -2.262$ or $t > 2.262$

Conclusion: Since the calculated value of t falls in the critical region, we may reject null hypothesis. That is, there is significant evidence to conclude that sales values of the companies between 2014 and 2015 differ significantly.

Exercise: Memory capacity of the students were tested before and after training given to the students. State whether the training was effective or not from the following scores? Consider $\alpha = 0.05$ to test the hypothesis.

Students #	1	2	3	4	5	6	7	8	9
Score before Training	12	9	15	19	13	18	7	10	17
Score after Training	5	23	20	13	8	21	8	16	12

[Hint: $H_0: \mu_x - \mu_y = 0$ or, $H_0: \mu_d = 0$ against $H_1: \mu_d > 0$; $\mu_x = \text{mean score after training}$, $\mu_y = \text{mean score before training}$].

Correlation Test

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Hypothesis for Correlation Test

$$H_0 : \rho = 0 \quad (\text{no correlation})$$

$$H_1 : \rho \neq 0 \quad (\text{significant correlation})$$

Where ρ is the population correlation coefficient.

The **test statistic** is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

- n = sample size
- Degrees of freedom $df = n - 2$
- Compare t with the critical t-value at desired significance level.

Example 1. A chemical engineer records the reaction rate at 7 different temperatures:

Temperature (°C)	Rate (mol/L·s)
30	1.2
35	1.5
40	1.7
45	2.0
50	2.2
55	2.5
60	2.8

Test at 5% level of significance whether temperature and reaction rate are significantly related.

Step 1: Compute correlation coefficient r

After calculation (omitting lengthy arithmetic):

$$r = 0.995$$

Step 2: Hypotheses

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

Step 3: Test statistic

$$t = \frac{0.995\sqrt{7-2}}{\sqrt{1-0.995^2}}$$

$$t = \frac{2.226}{0.1} = 22.26$$

Step 4: Critical value

- $df = 5$
- Two-tailed, $\alpha = 0.05 \rightarrow t_{0.025,5} = 2.571$

Step 5: Decision

- $22.26 > 2.571 \rightarrow \text{reject } H_0$

Conclusion: There is a significant positive correlation between temperature and reaction rate.

Example 2. A student measures voltage and current in 6 trials:

Voltage (V)	Current (A)
10	2.1
12	2.5
14	2.8
16	3.1
18	3.5
20	3.9

Test at 1% level of significance whether voltage and current are significantly related.

Proportion Test

1. One-Sample Proportion Test

Used when we compare a sample proportion $\hat{p}$ with a known population proportion p_0 .

Test statistic (Z):

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- n = sample size
- Compare Z with critical Z at desired significance level.

Hypotheses:

- Null: $H_0 : p = p_0$
- Alternative: $H_1 : p \neq p_0$ (two-tailed)

Example 1. A chemical plant produces 500 valves per batch. Historically, 4% of valves are defective. In a sample of 100 valves, 8 were found defective. Test at 5% level whether the defect rate has changed.

Solution:

Step 1: Sample proportion

$$\hat{p} = 8/100 = 0.08$$

Step 2: Hypotheses

$$H_0 : p = 0.04$$

$$H_1 : p \neq 0.04$$

Step 3: Test statistic

$$Z = \frac{0.08 - 0.04}{\sqrt{0.04 \times 0.96/100}}$$

$$Z = 2.04$$

Step 4: Critical value

- Two-tailed, 5% $\rightarrow Z = \pm 1.96$

Step 5: Decision

$$2.04 > 1.96 \rightarrow \text{reject } H_0$$

Conclusion: The defect rate is significantly higher than 4%.

2. Two-Sample Proportion Test

Used to compare two sample proportions $\hat{p}_1$ and $\hat{p}_2$.

Test statistic (Z):

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Where $\hat{p}$ = pooled proportion:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

- x_1, x_2 = number of successes in each sample

Example 2. Two LED manufacturers supply batches to a factory. Manufacturer A: 200 LEDs tested, 12 were faulty, and Manufacturer B: 150 LEDs tested, 3 were faulty. Test at 5% level whether the proportion of faulty LEDs differs between the two manufacturers.

Solution:

Hypothesis, $H_0: P_1 = P_2$ against $H_1: P_1 \neq P_2$

Step 1: Sample proportions

$$\hat{p}_1 = 12/200 = 0.06$$

$$\hat{p}_2 = 3/150 = 0.02$$

Step 2: Pooled proportion

$$\hat{p} = \frac{12 + 3}{200 + 150} = 15/350 \approx 0.04286$$

Step 3: Standard error

$$\begin{aligned} SE &= \sqrt{0.04286 \times (1 - 0.04286) \left(\frac{1}{200} + \frac{1}{150} \right)} \\ &= \sqrt{0.0411 (0.005 + 0.00667)} \\ &= \sqrt{0.0411 \times 0.01167} = \sqrt{0.000479} \approx 0.0219 \end{aligned}$$

Step 4: Z statistic

$$Z = \frac{0.06 - 0.02}{0.0219} \approx \frac{0.04}{0.0219} \approx 1.83$$

Step 5: Critical value

- Two-tailed, 5% $\rightarrow \pm 1.96$

Decision: $1.83 < 1.96 \rightarrow$ fail to reject H_0

Conclusion: No significant difference in faulty proportion between manufacturers.

Exercise

1. A laboratory reports that 5% of samples fail quality test under standard reactor conditions. In a batch of 120 samples, 10 failed. Test at 5% level whether the failure proportion has increased.
2. Two suppliers provide capacitors: Supplier A: 150 capacitors, 9 failed, and Supplier B: 130 capacitors, 4 failed. Test at 5% level whether the failure rates differ between the two suppliers.
3. Historically, 2% of filters are clogged in a production process. A sample of 80 filters shows 4 clogged filters. Test at 5% level whether the clogging rate has increased.
4. Two production lines make microchips: Line 1: 100 chips, 5 defective, and Line 2: 120 chips, 2 defective. Test at 1% level whether the defect proportions differ.

Contingency table- χ^2 test

The table which shows the arrangement of qualitative variables is known as contingency table. They deal with two or more nominal or ordinal variables where the data consist of a frequency count. In this table it is important that the categories of each attributes are mutually exclusive and exhaustive.

General case ($r \times p$ contingency table):

Suppose A and B are two attributes where A has r mutually exclusive and exhaustive categories $A_1, A_2, \dots, A_r$ and B has p mutually exclusive and exhaustive categories $B_1, B_2, \dots, B_p$. The table below indicates the general form of a contingency table,

Categories of B Categories of A	1	2	...	J	...	p	Total
1	O_{11}	O_{12}	...	O_{1j}	...	O_{1p}	R_1
2	O_{21}	O_{22}	...	O_{2j}	...	O_{2p}	R_2
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
i	O_{i1}	O_{i2}		O_{ij}		O_{ip}	R_i
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
r	O_{r1}	O_{r2}	...	O_{rj}	...	O_{rp}	R_r
Total	C_1	C_2	...	C_j	...	C_p	N

Where, O_{ij} = Observed cell frequency of $(i,j)^{th}$ cell/class,

R_i = Total of i 'th category of attribute A,

C_j = Total of j 'th category of attribute B

$$\text{And } N = \sum_{i=1}^r R_i = \sum_{j=1}^p C_j$$

Now calculate the expected cell frequency of $(i,j)^{th}$ cell/class E_{ij} as,

$$E_{ij} = \frac{R_i \times C_j}{N}$$

Here we want to test,

H_0 = There is no association between A and B

Against H_1 = There is association between A and B.

Assumption:

In calculating expected cell frequencies, we assume that the row and column totals are fixed.

To test H_0 , the test statistic is

$$\begin{aligned} \chi^2 &= \sum_{i=1}^r \sum_{j=1}^p \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \text{ which follows } \chi^2 \text{ distribution with } (r-1)(p-1) \text{ d.f.} \\ &= \sum_{i=1}^r \sum_{j=1}^p \frac{O_{ij}^2}{E_{ij}} - N \end{aligned}$$

Let the level of significance is α . The test is a one tailed test. The critical value of χ^2 is $\chi^2_{\alpha, (r-1)(p-1)}$

Decision: If $\chi^2 \geq \chi^2_{\alpha, (r-1)(p-1)}$, we can reject H_0 . Otherwise accept H_0 .

Problem 1: A random sample of 200 employees were drawn from a business enterprise to see if there is a relationship between educational attainment and job performance. The data on this appear below. Do you find any relationship between educational attainment and job performance of these employees? Test at 5% level of significance.

Job performance	Educational attainment			Total
	Below primary	College	University	
Excellent	10	40	10	60
Good	30	30	20	80
Fair	10	30	20	60
Total	50	100	50	200

Soultion:

H_0 : There is no association between the educational attainment and job performance

Against H_1 : There exists association between the educational attainment and job performance

Job performance	Educational attainment			Total
	Below primary	College	University	
Excellent	10 (15)	40 (30)	10 (15)	60(R_1)
Good	30 (20)	30 (40)	20 (20)	80(R_2)

Fair	10 (15)	30 (30)	20 (15)	60(R_3)
Total	50(C_1)	100(C_2)	50(C_3)	200(N)

To calculate expected cell frequency:

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{60 \times 50}{200} = 15, E_{21} = \frac{R_2 \times C_1}{N} = \frac{80 \times 50}{200} = 20, E_{32} = \frac{R_3 \times C_2}{N} = \frac{60 \times 100}{200} =$$

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The χ^2 is calculated as follows

(Row, Column)	O_{ij}	E_{ij}	$O_{ij} - E_{ij}$	$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
(1,1)	10	15	-5	1.67
(1,2)	40	30	10	3.33
(1,3)	10	15	-5	1.67
(2,1)	30	20	10	5.00
(2,2)	30	40	-10	2.50
(2,3)	20	20	0	0
(3,1)	10	15	-5	1.67
(3,2)	30	30	0	0.00
(3,3)	20	15	5	1.67
Total	200	200	0	17.51(χ^2)

Level of significance, $\alpha = 0.05$

The critical value of χ^2 is $\chi^2_{\alpha, (r-1)(p-1)} = \chi^2_{0.05, 4} = 9.49$

Decision: Since $\chi^2 \geq 9.49$, we can reject H_0 . We can conclude that there exists association between educational attainment of the employees and their job performance.

Exercise 1: To study if there is any association between the gender of a student and the major that he/she has as their first choice in science division, a random sample of 141 students had the following responses:

Gender	Male	Female
Mathematics	12	29
Biology	27	16
Physics	15	19
Chemistry	9	14

Use a 5% level of significance and determine if there is any association between gender and the major.

Exercise 2: A sample of 500 children was classified according to their mothers' level of education: illiterate, primary, secondary and higher secondary. An immunization campaign was initiated by an NGO. At the end of the campaign the following Statistics were obtained:

	Illiterate	Primary	Secondary	Higher	Total
Immunized	150	100	80	70	400
Not Immunized	40	30	20	10	100
Total	190	130	100	80	500

Do the rates of immunization differ significantly from one level of education to another? Use 5% level of significance.

2 × 2 Contingency table: The contingency table with only two rows and two columns is called 2 × 2 contingency table. Let in the table the attribute A has two mutually exclusive and exhaustive categories A_1 and A_2 and the attribute B has two mutually exclusive and exhaustive categories B_1 and B_2 . The table is given below

2 × 2 Contingency table

B A	B_1	B_2	$Total$
A_1	a	b	$R_1=a+b$
A_2	c	d	$R_2=c+d$
$Total$	$C_1=a+c$	$C_2=b+d$	$N=a+b+c+d$

We are to test

H_0 : There is no association between A and B

Against, H_1 : There is association between A and B .

Test statistic, $\chi^2 = \frac{N(ad-bc-\frac{N}{2})^2}{(a+b)(a+c)(c+d)(b+d)}$ which follows χ^2 distribution with 1 d.f.

Let the level of significance is α . The test is a one tailed test. The critical value of χ^2 is $\chi^2_{\alpha,1}$.

Decision: If $\chi^2 \geq \chi^2_{\alpha,1}$, we can reject H_0 .

Problem 1. A study records whether people smoke and whether they have lung disease.

	Lung Disease	No Lung Disease	Total
Smoker	36	14	50
Non-Smoker	18	32	50
Total	54	46	100

Is lung disease independent of smoking?

Problem 2. A class compared pass rates for students who used an online support course.

	Passed	Failed	Total
Used Course	42	18	60
Did Not Use	30	30	60
Total	72	48	120

Is course usage associated with exam success?

Problem 3. A company studied whether job choice depends on gender.

	Technical Job	Non-Technical Job	Total
Male	40	20	60
Female	25	35	60
Total	65	55	120

Is job type independent of gender?